Computer Communication and Networks

Physical Layer

Session 6

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The Physical Layer

Learning Outcome:- At the end of the lesson, you are expected to be able to:

- Define unguided media
- Identify wireless communication category



The Transmission Media: Unguided

Unguided Media:

- It is also known as Wireless Communication
- Transport the **electromagnetic waves** without using a physical conductor.
- Signals are **broadcast through air** (few cases in water) and thus available to anyone who has a device capable of receiving them.
- The section of the electromagnetic spectrum defined as **radio communication is divided into eight ranges, called bands.**
- Bands are rated from very low frequency(VLF) to extremely high frequency(EHF).



The Transmission Media: Unguided

- Eight Bands are shown below:

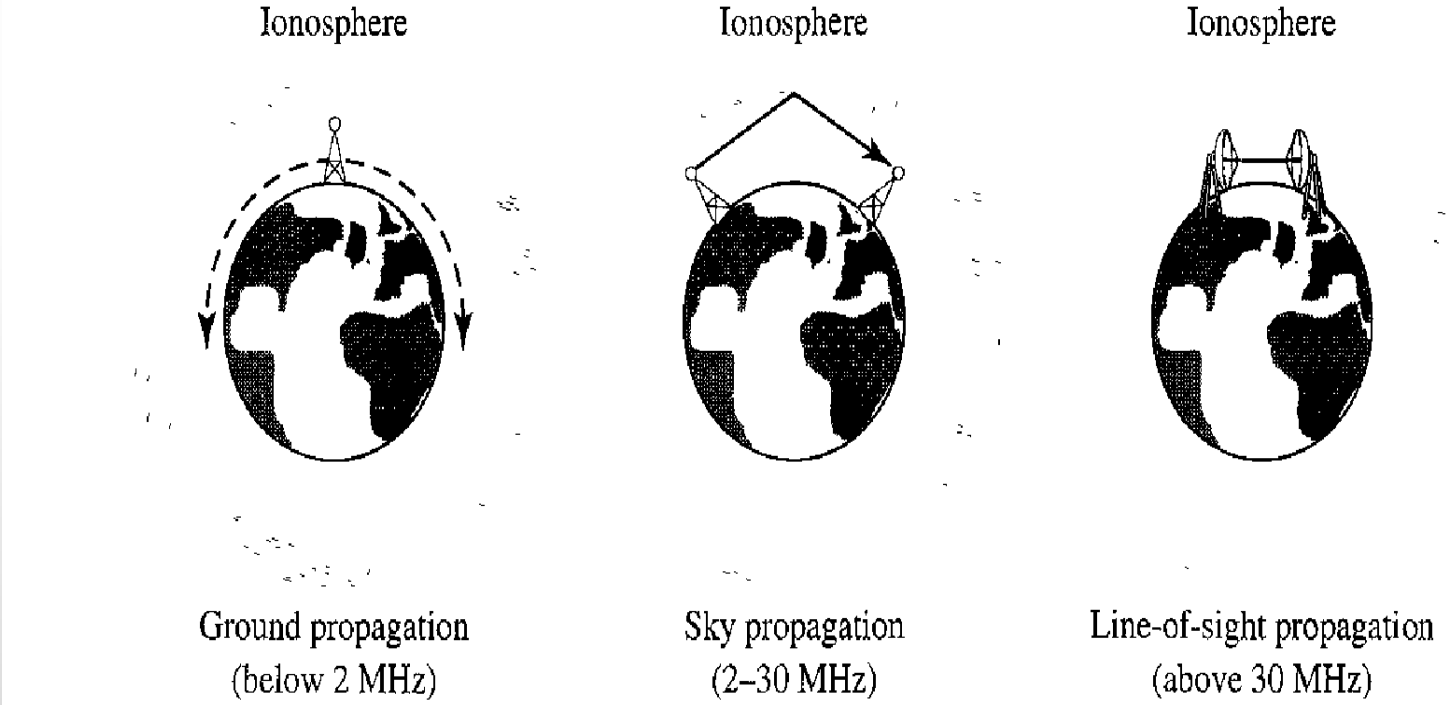
Band	*Range*	*Propagation*	*Application*
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite



The Transmission Media: Unguided

- The 3 propagations of unguided media:
  - Unguided signals can travels from source to destination in several ways such as:

1. Ground propagation
2. Sky propagation
3. Line-of-sight propagation.





The Transmission Media: Unguided

- Wireless communication: Wireless Communication can be divided into 3 broad groups:

1. **Radio Waves**
   - Frequency range: 3kHz – 1 GHz
   - Omni directional
   - Long distance broadcasting such as AM radio.
   - Applications: AM & FM radio, television, maritime radio, cordless phone, etc. for multicast communications.



The Transmission Media: Unguided

- Wireless communication: Wireless Communication can be divided into 3 broad groups:

2. Micro Waves

- Frequency range: 1 – 300 GHz.
- Unidirectional: Sending and receiving antenna needs to be aligned.
- Unicast(one-to-one) communication
- Applications: Cellular phones, satellite networks, & wireless LANs for unicast communication



The Transmission Media: Unguided

- Wireless communication: Wireless Communication can be divided into 3 broad groups:

3. Infrared

- Frequency: 300GHz – 400 THz.
- Infrared signals are used for short-range communications in a closed area.
- Applications: Used in communicating Keyboard, mice, PCs & Printers



References

References:

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2. Tanenbaum, Computers Networks, 5th Edition.
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4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan